Statement of Purpose

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Research Interest and Motivation. I come from Africa where access to cutting-edge technology to solve healthcare challenges remains a luxury for most people but also for health institutions. For example, most hospitals in sub-Saharan Africa do not have MRI (Magnetic Resonance Imaging) units for patient screening and high-precision diagnostics. Despite the precarious state of most African healthcare systems, the global boom in mobile technology and artificial intelligence (AI) presents a great opportunity for the Global South to develop cost-effective, cutting-edge healthcare solutions and address societal issues with efficiency and impact. My passion for building intelligent technologies and the healthcare challenges faced by community motivate my strong interest in efficient and private medical AI research – in particular in resource-scarce environments – to improve healthcare systems and bring meaningful benefits to society.

Medical AI and Federated Learning. After completing the Introduction to Deep Learning course, I was inspired to develop AI-driven healthcare solutions and conduct impactful research that could save lives in Africa. Building on this course project, I created and set up the HumekaFL health AI research project with Professor Carlee Joe-Wong, which is currently ongoing at Carnegie Mellon University (CMU) Africa[1]. HumekaFL is a cost-effective, user-friendly AI mobile app to help medical professionals diagnose birth asphyxia in newborns accurately and timely[2].

As the project lead, I developed a Support Vector Machine (SVM) model using baby cries within a cross-silo federated simulation environment. Federated Learning (FL) allows multiple clients, like hospitals in HumekaFL, to collaboratively train a shared machine learning model under a central server's coordination. Clients update the model locally, send updates to the server, and receive an aggregated global model, continuing this cycle until convergence. Our federated learning SVM (FedSVM) model outperformed many state-of-the-art centralized neural models for classifying birth asphyxia (BA). We chose FedSVM due to our small training dataset and to ensure the privacy and security of sensitive health data while leveraging collaborative learning in resource-constrained settings like African hospitals. To address privacy concerns, we incorporated a differential privacy layer, which protects individual data by adding calibrated noise, ensuring analysis outcomes remain consistent regardless of individual data inclusion. Our initial findings [2], were presented at the ACM Compass conference as a poster paper this year and I am the first author. The model has been integrated into a mobile application for deployment in health institutions. While our research demonstrates good model performance, our dataset lacks cries from African babies. To address this, we have initiated discussions with health institutions in Kenya and Malawi to collect relevant data.

HumekaFL sparked my interest in using **federated and multimodal ML** models to enhance biomedical imaging, particularly for resource-constrained medical devices. I am especially interested in **Vision Language Models** (VLMs), which combine image and text data for medical imaging applications. In healthcare, MRI and X-rays are crucial for diagnosing diseases such as arthritis and cancer. However, uncertainties in scanned images can lead to misdiagnosis and delayed treatments, especially in resource-limited settings where advanced imaging tools are often unavailable. I believe that leveraging federated VLMs for low-resource medical devices could revolutionize **ML-aided diagnosis and personalized care**. This technology could make advanced healthcare accessible not only to patients in well-equipped hospitals in the Global North but also in **resource-limited settings** in the Global South. However, applying these models to healthcare data raises two significant challenges: (1) Resource Constraints: Medical devices and wearables in low-resource settings may lack the capacity to run VLMs effectively. (2)Adversarial Robustness: Resource-scarce and distributed environments introduce unique **adversarial robustness** challenges. In ML, this refers to a model's ability to resist attacks like model poisoning and data recovery attacks (e.g., gradient inversion). Due to hardware limitations, these devices cannot effectively filter adversarial inputs as computationally heavy devices can. Additionally, adversarial inputs from one device could compromise others in a federated training setup. These challenges will shape my research directions as I pursue a PhD program, focusing on developing solutions that ensure the effective and safe deployment of federated VLMs in healthcare settings.

Human-Robot Interaction. My research project to complete the advanced study in the ECE program was titled Culturally-Sensitive Human-Robot Interaction: A Case Study with the Pepper Humanoid Robot. Under the guidance of Professor David Vernon, I developed robot behaviors tailored for effective human-robot interaction within African social contexts. My research highlighted that trust is essential

[1] HumekaFL is a joint effort between the Lions Research Group and the Upanzi Network.

[2] BA is the inability to breathe and one of the main causes of neonatal death globally.

not only for successful human interactions but also for the acceptance of robots as social entities, emphasizing that trust is culture-dependent. By creating culture-aware robot behaviors, humanoids are more likely to be accepted in society. My findings were showcased during a university lab tour at the CMU-Africa campus. After developing a Robot Operating System (ROS) bridge to control the Pepper robot's actuators and sensors, I used African cultural knowledge to develop behaviors enabling the robot to welcome guests and respectfully greet them by bowing and showing objects without pointing at them. Following this, I conducted a Robotic Social Attributes Scale (RoSAS) evaluation survey, where participants rated the robot as compassionate, interactive, sociable, competent, and reliable. This research resulted in two publications [3, 4] and served as the foundation for the research project Culturally Sensitive Social Robotics in Africa. Additionally, I received the Jeremiah N. Mpagazehe Rising Researcher Award for my research contributions. As a PhD student, I find the intersection of HRI and medical AI to be an exciting area for research.

Teaching Assistance and Extra-Curricular Activities. As a graduate student at CMU, my research learning pathway was also consolidated by gaining teaching experience in the following courses: Robotics Principles and Practices and Cognitive and Cognitive Robotics. As a teaching assistant, I helped students understand better concepts taught in class, prepare lab resources (e.g. calibrating the Roomba robot) to get their assignments done.

At CMU-Africa, I led the IoT Club. My leadership transformed the club into one of the most successful on campus. We increased student engagement through project-based sessions, organized visits to companies like Zipline, and hosted industry workshops. Recognizing the need for greater research exposure, I co-founded the Research Club, which provided tutorials and workshops with faculty at CMU-Africa, helping students learn about ongoing research and essential skills for their careers. My leadership efforts earned me a Women in Engineering–IDEA travel grant to attend the International Conference in Robotics and Automation (ICRA), and I was featured in the IEEE Robotics and Automation Magazine [1].

Conclusion. I am strongly motivated to pursue my PhD at the University of California San Diego. In particular, I would love to work with Professor **Edward Wang** whose research at the intersection mobile health sensing, signal processing, wearable computing, human computer interaction and more, align with my research interests. Professor **Michael Yip**'s research in reinforcement learning for robots, computationally efficient robot planning, autonomous robotic surgery also resonates with my research aspirations. Furthermore, I will also be honored to work with Professor **Farinaz Koushanfar** whose research also touches medical devices, automotive systems, deep learning networks and secure bioinformatics. Completing my PhD at the University of California San Diego will not only advance my career in medical AI but also help me create meaningful global impact. I look forward to joining the University of California San Diego as a PhD student in Fall 2025.

References

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